



Asset Performance:

Hea 10 Carbon monoxide detection



Credits



No Minimum Standard

Aim

To protect asset users from harmful levels of carbon monoxide associated with the asset's combustion appliances and enclosed parking areas.

Question

Are carbon monoxide detection and alarm systems installed in the asset in spaces that contain combustion appliances and in enclosed parking areas?

Credits	Answer	Select all answers that apply
0	A.	Question not answered
0	B.	No
1	C.	Yes, in all spaces containing combustion appliances
1	D.	Yes, in all enclosed parking areas

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Filtering Where there are no combustion appliances or enclosed parking areas present in the asset, the associated credits can be filtered out of the assessment.	C, D
2.	The carbon monoxide detection system must alert asset users to acute levels of carbon monoxide, therefore enabling users to react before being exposed to a significant risk (rather than monitoring low levels for health purposes only). A detection system should consist of: a) A carbon monoxide detector in every space containing a combustion appliance, e.g. those used for space and water heating and cooking. b) A carbon monoxide detector in every space where a combustion appliance's flue passes through rooms or space used as sleeping accommodation.	C, D

Criterion	Assessment criteria	Applicable Answer
	c) Carbon monoxide detectors in the asset's enclosed parking areas (where these are within scope of the assessment), e.g. underground car parks.	
3.	Carbon monoxide detector/alarm systems must be permanently installed. They may be self-contained detector and alarm units powered by a battery designed to operate for the working life of the detector or they may be connected to the asset's electricity supply, either with a built-in sounder or connected to a central control panel. The detector should incorporate a warning device to alert asset users when the working life of the detector is due to expire or, if connected to the electricity supply, fitted with a sensor failure warning device.	C, D
4.	Carbon monoxide detection systems must be installed and maintained in accordance with the manufacturer's instructions and must be suitable for application in the relevant asset type and space. Detectors intended for application in domestic or residential assets may be used where manufacturer's instructions state that the detectors can also be used for commercial application within the asset type under assessment.	C, D
5.	Carbon monoxide detectors that are in spaces or areas that asset users will not normally frequent, for example, those located within boiler rooms or car parks, should be linked to a visual or audible alarm or control panel sited at a staffed location, such as a reception desk.	C, D
6.	The provision of a carbon monoxide detection system should not be regarded as a substitute for the correct installation and regular servicing of a combustion appliance.	C

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
2 – 6	Photographic evidence of the detection and alarm system(s).
2 – 6	Specifications, procedures or monitoring logs for the detection and alarm system(s).
2 – 6	Diagrams, photographs or plans indicating location of combustion appliances and enclosed car parks and associated carbon monoxide detectors.
2 – 6	Confirmation from the asset owner or management team where no combustion appliances are installed in the asset or where there are no enclosed parking areas.

Definitions

Carbon monoxide:

Carbon monoxide is a colourless, odourless, and tasteless gas. Low levels of carbon monoxide gas can be present in the atmosphere; however, it is highly toxic and dangerous to humans and animals in higher quantities. The gas is produced in high levels from appliances where incomplete combustion of a carbon-based fuel occurs. Incomplete combustion could occur in appliance installations that are defective, lack proper maintenance or have inadequate provision for combustion air.